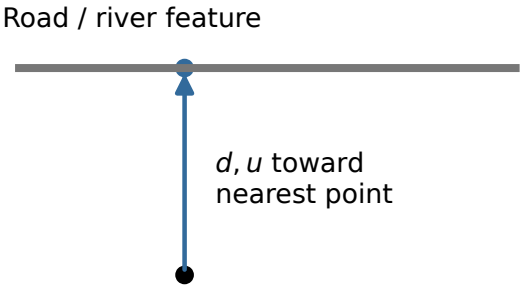
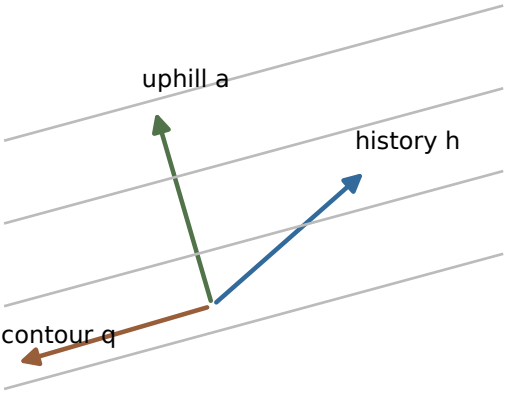
(a) Nearest-point geometry



Not a tangent or flow direction

(b) Signed projections



$h^T a$, $h^T q$: no slope multiplier

(c) Frozen WorldCover groups and road interaction

built	vegetation	water/wetland	valid other
50	10,20,30,40, 95,100	80,90	other valid codes

$1_{\text{group}} \log(1 + d/(1\,m))$; nodata is missing, not valid other.